

BIOTECH PRODUCTS



Bt Corn

resistant to Asian corn borer
Ostrinia furnacalis
Guenee



Biofertilizers

natural fertilizers that consist of live biomass or dormant cells of effective microbial strains



Malusog Rice

genetically-engineered crop to alleviate vitamin A deficiency



Submergence-Tolerant Rice

high-yielding rice capable of enduring transient complete submergence



ELISA kits

DNA-based detection kits for crop diseases

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DA-Crop Biotechnology Center



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Innovation and Collaboration
for High-Impact Biotech R4D

DA-CROP BIOTECHNOLOGY CENTER

The **Crop Biotechnology Center (CBC)** is one of three biotechnology centers under the Department of Agriculture (DA). It serves as a catalyst in the development and utilization of modern biotechnologies for agriculture. It rationalizes and implements the research, development, and extension agenda of the DA-Biotech Program. It also provides biotechnology capacity-building services to a network of R4D institutions in the use of modern crop biotechnologies.

DA-CBC focuses on applying advanced technologies that contribute to the improvement in performance, yield, and quality of important commodities, such as rice, corn, coconut, and high value crops, such as mango, garlic, onion, eggplant, tomato, cotton, rubber, cassava, sweet potato, yam, cacao, papaya, banana, and abaca.

R4D AREAS

- **Genetics**
- **Genetic Engineering and Gene Editing**
- **Tissue Culture**
- **Microbial Biotechnology**
- **Diagnostics**
- **Bioinformatics and Computational Biology**
- **Natural Products, Natural Ingredients, and Value-Adding**

R4D FACILITIES:



LABORATORIES

Molecular Genetics

- Functional genomic approaches to accelerate crop breeding and production
- Construction of detailed, high-resolution QTL maps and physical maps for map-based cloning

Genetic Transformation

- Use of genetic manipulation to improve traits of interest
- Development of genetically modified (GM) and non-GM organisms, alterations to gene sequences, and gene knock-outs (RNAi)

Tissue Culture

- Aseptic propagation of plant parts in a highly controlled environment
- Creation of genetic variability for crop improvement

Systems Biology ('omics')

- Systems biology approaches combining metabolomics with proteomics and/or transcriptomics
- Experimental strategies to characterize ribonucleic acid-protein complexes/interactions

Microbial Biotechnology

- Use of plant-growth-promoting microorganisms (PGPM) as biofertilizers to enhance plant and soil health and biopesticides to control invasive plants and insects

Bioinformatics and Computational Biology

- Use of bioinformatics tools and high-performance computing systems and various computing support services to transform complex data into information useful to scientists

Molecular Diagnostic

- Development of molecular tests, including biochemical, electrochemical, optical, and imaging-based, to diagnose diseases or other pathological conditions

TRAINING/ CONFERENCE VENUES

Plenary Hall	(Capacity: 600 pax)
Multipurpose Hall	(Capacity: 300 pax)
Training Rooms	(Capacity: 300 pax)